# XML cleanup and its preservation guards

Abbreviated source examples, negative cases and native rendering evidence

This atlas follows the current Default policy. It groups the reviewed behavior specification, implementation, commit history and focused regression cases into 28 visual examples. Experimental Maximum and Too Aggressive settings have different preservation policies.

Read each source/result pair at its native one-column width. Red backgrounds and minus signs identify removed lines. Green backgrounds and plus signs identify added lines. An equals sign identifies a protected structure. Folded color bands connect edited ranges to thin markers at their exact gaps. These cues supplement syntax color.

The XML is illustrative. Namespace wrappers, irrelevant attributes, cached payloads and repeated cells are omitted. Style indices have meaning only within their own example. Comments describe omitted scope or an exclusion rather than literal output. The complete native font-control packages are retained separately.

The source index pins each case to a file and a regression-test or implementation anchor. A matching unit test checks its own fixture. It does not prove that an abbreviated picture is a complete runnable workbook or that every possible Excel presentation is preserved.

Native evidence. The new reduced font control has three real Excel captures after CalculateFullRebuild. The retained Enron fill comparison has three historical captures after rebuild and Save. All other outcome descriptions are traced to their named tests or investigation records. No schematic is presented as an Excel screenshot.

Boundaries. Unsupported XML can exclude an input. Unknown inheritance or references follow the local conservative guard. Ordinary hidden tables are not covered by hidden-tail admission. Alias-name cleanup does not currently prove equality of formula bodies. Selected-region native equality is not whole-workbook certification.

Reviewed source revision e6ae437a2d72d1d26ede5e1f272b53d06fc3660c. Core verification completed with 298 passed, zero failed and one manual timing test ignored. The source repository was read and tested for this figure task. Cleanup behavior was not edited.

Design references: Animated Vega-Lite Figure 2 and Bluefish Figure 2. This atlas uses newly drawn geometry, editable text, local diff highlighting and native control images.

# Guarded XML transformations

Each panel is 252 pt wide. Code is 8.2 pt at this scale.

## 01 Content is not formatting

**Source**

```
<row r="1">
 <c r="A1"><v>0</v></c>
 <c r="B1"><f t="shared" si="0"/></c>
 <c r="C1">
  <v xml:space="preserve">7</v></c>
</row>
```

=

**Retained by the guard**

```
<row r="1">
 <c r="A1"><v>0</v></c>
 <c r="B1"><f t="shared" si="0"/></c>
 <c r="C1">
  <v xml:space="preserve">7</v></c>
</row>
```

Keep zero, shared-formula followers and attributed value elements. Empty cached display is insufficient.

Unsafe shortcut. Deleting a zero or a formula changes workbook data.

Abbreviated XML. Style IDs are local.

## 02 Keep one font witness

**Source**

```
 <row r="2">
  <c r="A2">...</c>
  <c r="B2" s="1"/>
- <c r="C2" s="1"/>
 </row>
```

**Guarded result**

```
 <row r="2">
  <c r="A2">...</c>
  <c r="B2" s="1"/>
 </row>
```

A2 uses Arial 10. B2 and C2 carry Arial 18. Keep one needed font representative in this row.

Unsafe shortcut. Removing both witnesses shortens this controlled Excel row from 23.25 to 12.75 points.

Abbreviated XML. Style IDs are local.

## 03 A witness is local to a merge

**Source**

```
 <row r="6" ht="177"
  customHeight="1">
  <c r="A6" s="237">...</c>
  <c r="B6" s="238"/>
- <c r="C6" s="238"/>
- <c r="H6" s="238"/>
 </row>
 <mergeCell ref="A6:H6"/>
```

**Guarded result**

```
 <row r="6" ht="177"
  customHeight="1">
  <c r="A6" s="237">...</c>
  <c r="B6" s="238"/>
 </row>
 <mergeCell ref="A6:H6"/>
```

Retain one representative per encountered font in each merged area and row, even at fixed height.

Unsafe shortcut. Deleting B6:H6 changed justified paragraph spacing. A witness in another merge is insufficient.

Abbreviated XML. Style IDs are local.

## 04 No fill can be a visible override

**Source**

```
<row r="6" s="2"
 customFormat="1" ht="18"
 customHeight="1">
 <c r="K6"/>
 <c r="L6"/>
</row>
```

=

**Retained by the guard**

```
<row r="6" s="2"
 customFormat="1" ht="18"
 customHeight="1">
 <c r="K6"/>
 <c r="L6"/>
</row>
```

Row style 2 has a blue fill. The explicit styleless cells clear it. Keep every local mask, not one font representative.

Unsafe shortcut. Deleting K6 exposes blue at K6, even if L6 remains.

Abbreviated XML. Style IDs are local.

# Guarded XML transformations

Each panel is 252 pt wide. Code is 8.2 pt at this scale.

## 05 Remove an override with nothing to clear

**Source**

```
- <row r="2" ht="12.75"
-  customHeight="1">
-  <c r="A2" s="5"/>
-  <c r="B2" s="5"/>
- </row>
```

**Guarded result**

```
  <!-- Content-free tail removed -->
```

Style 5 applies fillId=0. No visible inherited fill or border exists and no other retention rule applies.

Unsafe shortcut. The same attributes must survive when a white or colored parent fill is present.

Abbreviated XML. Style IDs are local.

## 06 An implicit row style still overrides

**Source**

```
<col min="6" max="6" style="17"/>
<row r="187" customFormat="1">
 <c r="F187" s="17"/>
</row>
```

=

**Retained by the guard**

```
<col min="6" max="6" style="17"/>
<row r="187" customFormat="1">
 <c r="F187" s="17"/>
</row>
```

customFormat=1 without s applies row XF 0. A gray cell may restore a column fill that the row format cancels.

Unsafe shortcut. Removing F187 because it matches the column exposes the row no-fill format.

Abbreviated XML. Style IDs are local.

## 07 Font witnesses cannot replace local masks

**Source**

```
<col min="6" max="8"
 hidden="1" style="13"/>
<row r="14">
 <c r="A14" s="26">...</c>
 <c r="F14" s="26"/>
 <c r="G14" s="26"/>
</row>
```

=

**Retained by the guard**

```
<col min="6" max="8"
 hidden="1" style="13"/>
<row r="14">
 <c r="A14" s="26">...</c>
 <c r="F14" s="26"/>
 <c r="G14" s="26"/>
</row>
```

Hidden columns can affect automatic height. Preserve each mask that would expose a font absent from the original row.

Unsafe shortcut. A matching visible witness cannot stop a larger inherited font at F14 from affecting the row.

Abbreviated XML. Style IDs are local.

## 08 Move repeated formatting into the row

**Source**

```
  <row r="1" ht="9"
-  customHeight="1">
    <c r="A1" s="3"><v>7</v></c>
-  <c r="B1" s="2"/>
    <!-- repeated through XFD1 -->
-  <c r="XFD1" s="2"/>
  </row>
```

**Guarded result**

```
  <row r="1" ht="9"
+  customHeight="1"
+  s="2" customFormat="1">
    <c r="A1" s="3"><v>7</v></c>
  </row>
```

Reuse an existing XF. Require a visible fixed-height, unprotected row and proved inheritance. Dense route needs at least 1,024 occurrences.

Unsafe shortcut. A newly inherited fill, border, protection or font at a gap can change the workbook.

Abbreviated XML. Style IDs are local.

# Guarded XML transformations

Each panel is 252 pt wide. Code is 8.2 pt at this scale.

## 09 Restore necessary exceptions at gaps

**Source**

```
  <row r="1" ht="9"
-  customHeight="1">
   <c r="A1" s="3"><v>7</v></c>
   <!-- repeated s=2 cells -->
   <!-- E1 was absent -->
  </row>
```

**Guarded result**

```
  <row r="1" ht="9"
+  customHeight="1"
+  s="2" customFormat="1">
   <c r="A1" s="3"><v>7</v></c>
+  <c r="E1" s="3"/>
  </row>
```

The bounded dense-gap route admits at most 64 missing columns and restores original formats only when the inheritance proof succeeds.

Unsafe shortcut. Promoting the row without an E1 exception exposes a new inherited font or fill.

Abbreviated XML. Style IDs are local.

## 10 A cached height is not a fixed height

**Source**

```
<row r="105">
 <c r="A105" s="2">...</c>
 <!-- many s=2 blanks -->
</row>
```

=

**Retained by the guard**

```
<row r="105">
 <c r="A105" s="2">...</c>
 <!-- many s=2 blanks -->
</row>
```

Do not promote an automatic row. Both ht and customHeight=1 are required, even when the proposed font is already present.

Unsafe shortcut. A historical promotion shortened Destatis row 105 from 12.75 to 9 points and moved the footer.

Abbreviated XML. Style IDs are local.

## 11 A centering span needs every blank

**Source**

```
<row r="2">
 <c r="D2" s="9">...</c>
 <c r="E2" s="9"/>
 <c r="F2" s="9"/>
 <c r="G2" s="9"/>
</row>
```

=

**Retained by the guard**

```
<row r="2">
 <c r="D2" s="9">...</c>
 <c r="E2" s="9"/>
 <c r="F2" s="9"/>
 <c r="G2" s="9"/>
</row>
```

Style 9 has horizontal=centerContinuous. A value or formula seeds a contiguous same-alignment span.

Unsafe shortcut. Keeping only G2 does not restore centering after F2 is deleted. A gap ends the span.

Abbreviated XML. Style IDs are local.

## 12 Protection does not erase a font effect

**Source**

```
<row r="2">
 <c r="A2">...</c>
 <c r="B2" s="7"/>
-<c r="C2" s="7"/>
</row>
```

**Guarded result**

```
<row r="2">
 <c r="A2">...</c>
 <c r="B2" s="7"/>
</row>
```

Style 7 contains a font and protection metadata. Outside the protection-retention area it may still supply required automatic height.

Unsafe shortcut. Treating protection-only eligibility as permission to remove its font can shift later content.

Abbreviated XML. Style IDs are local.

# Guarded XML transformations

Each panel is 252 pt wide. Code is 8.2 pt at this scale.

## 13 Prune only a genuinely redundant fill

**Source**

```
  <col min="1" max="16384" style="1"/>
  <row r="486"><c r="E486">...</c></row>
- <row r="487" s="2" customFormat="1"/>
  <!-- same tail through row 49633 -->
```

**Guarded result**

```
  <col min="1" max="16384" style="1"/>
  <row r="486"><c r="E486">...</c></row>
```

Column style 1 covers A:XFD with white. Tail row style 2 repeats that fill and has no remaining anchor. Check exact column overrides and gaps.

Unsafe shortcut. A blue tail on the same white columns remains visible. White fills also hide gridlines.

Abbreviated XML. Style IDs are local.

## 14 A font alone need not anchor a tail

**Source**

```
- <row r="1424" s="6"
-  customFormat="1">
-  <c r="A1424"/>
-  <c r="B1424" s="5"/>
- </row>
```

**Guarded result**

```
  <!-- Empty sheetData -->
```

There are no values, visible fills, borders, merges, drawings or later anchors. Font-only representatives do not retain an otherwise content-free sheet.

Unsafe shortcut. The same row before later visible content can be needed for layout and must be reconsidered in context.

Abbreviated XML. Style IDs are local.

## 15 A row header can clear column borders

**Source**

```
  <col min="6" max="6" style="4"/>
  <row r="2" s="7" customFormat="1"/>
```

=

**Retained by the guard**

```
  <col min="6" max="6" style="4"/>
  <row r="2" s="7" customFormat="1"/>
```

Column style 4 has a border. Row style 7 has explicit borderId=0. Keep the masking row header, including when applyBorder is absent.

Unsafe shortcut. Deleting the empty header exposes vertical column borders across that row.

Abbreviated XML. Style IDs are local.

## 16 Empty row heights position visible borders

**Source**

```
  <col min="6" max="6" style="4"/>
  <row r="20" ht="24"/>
  <row r="21" ht="12.75"/>
```

=

**Retained by the guard**

```
  <col min="6" max="6" style="4"/>
  <row r="20" ht="24"/>
  <row r="21" ht="12.75"/>
```

Visible column borders make stored heights of nonhidden tail rows observable. Preserve them even without customHeight.

Unsafe shortcut. Removing an empty height record changes the extent and segmentation of the visible borders.

Abbreviated XML. Style IDs are local.

# Guarded XML transformations

Each panel is 252 pt wide. Code is 8.2 pt at this scale.

## 17 Conditional ranges protect row layout

**Source**

```
  <row r="2" ht="17.25"
-  customHeight="1"><c r="H2" s="6"/></row>
- <row r="60" ht="20" customHeight="1"/>
  <conditionalFormatting sqref="A5:B44">
   ...
  </conditionalFormatting>
```

**Guarded result**

```
  <row r="2" ht="17.25"
+  customHeight="1"/>
  <conditionalFormatting sqref="A5:B44">
   ...
  </conditionalFormatting>
```

Keep layout through the conditional region. Remove unnecessary font-only cell payloads and unrelated later tail headers separately.

Unsafe shortcut. Deleting preceding row heights shifts the end of a conditional fill, even with no ordinary cell fill.

Abbreviated XML. Style IDs are local.

## 18 Objects extend the layout boundary

**Source**

```
<row r="1"><c r="A1"><v>7</v></c></row>
<row r="50" ht="30" customHeight="1"/>
<drawing r:id="rId1"/>
<!-- drawing anchor ends at row 60 -->
```

=

**Guarded result**

```
<row r="1"><c r="A1"><v>7</v></c></row>
<row r="50" ht="30" customHeight="1"/>
<drawing r:id="rId1"/>
```

Resolve supported drawing relationships and anchors before trimming row layout beyond the last data cell.

Unsafe shortcut. Deleting row 50 because A1 is the last value can move or resize the later object.

Abbreviated XML. Style IDs are local.

## 19 Merge tail columns with font witnesses

**Source**

```
  <col min="10" max="20" style="1"/>
- <col min="21" max="16384" style="2"/>
```

**Guarded result**

```
  <col min="10" max="20" style="1"/>
+ <col min="21" max="21" style="2"/>
+ <col min="22" max="16384" style="1"/>
```

This is a schematic equal-fill tail merge. Keep one original empty-tail column per font identity that replacement would otherwise erase.

Unsafe shortcut. Replacing every Calibri 11 column with Arial 10 shortened NLA data rows from 15.75 to 15 pt.

Abbreviated XML. Style IDs are local.

## 20 Hidden constants need deferred approval

**Source**

```
<row r="8" hidden="1">
 <!-- >=4096 contiguous integer 0s -->
 <c r="A8"><v>0</v></c>
 ...
 <c r="XFD8"><v>0</v></c>
</row>
```

**Decision**

```
<!-- Delete admitted suffix only after -->
<!-- reference and layout checks pass. -->
```

Admit only a dense horizontal suffix of equal signed integers ending at XFD, with every source cell eligible. Any intersecting known reference retains the entire span.

Unsafe shortcut. Arbitrary hidden tables, short runs, gaps, text, formulas and vertical repeats are not admitted.

Abbreviated XML. Style IDs are local.

# Guarded XML transformations

Each panel is 252 pt wide. Code is 8.2 pt at this scale.

## 21 Translate shared references by axis

**Source**

```
<c r="D166"><f t="shared" si="0"
 ref="D166:J166">D91/D64*100-100</f></c>
<c r="E166"><f t="shared" si="0"/></c>
<!-- hidden E64 = 4232.35 -->
```

=

**Retained by the guard**

```
<c r="D166"><f t="shared" si="0"
 ref="D166:J166">D91/D64*100-100</f></c>
<c r="E166"><f t="shared" si="0"/></c>
<!-- hidden E64 = 4232.35 -->
```

Translate relative axes over the follower range. Absolute axes remain fixed. Protect enclosing rectangles without enumerating every formula. In this example the protected input rectangles include D91:J91 and D64:J64.

Unsafe shortcut. Scanning only the anchor misses E64 and can produce DIV/0 errors after rebuilding.

Abbreviated XML. Style IDs are local.

## 22 A blank chart-source cell carries format

**Source**

```
<c r="A1" s="4"/>
<c r="A2"><v>0.25</v></c>
<!-- chart series references A1:A2 -->
<!-- axis format is source-linked -->
```

=

**Guarded result**

```
<c r="A1" s="4"/>
<c r="A2"><v>0.25</v></c>
```

Retain the first explicit cell of a directly resolved chart series, including its original number format. Do not retain every blank in the series.

Unsafe shortcut. Deleting A1 can change a source-linked chart axis even though A1 has no value.

Abbreviated XML. Style IDs are local.

## 23 Whitespace can block text overflow

**Source**

```
<c r="A1" t="inlineStr">
 <is><t>Long heading</t></is>
</c>
<c r="B1" t="s"><v>4</v></c>
<!-- shared string 4 is one space -->
```

=

**Guarded result**

```
<c r="A1" t="inlineStr">
 <is><t>Long heading</t></is>
</c>
<c r="B1" t="s"><v>4</v></c>
```

Default leaves optional whitespace-value deletion disabled. A space is a value and can stop neighboring text overflow.

Unsafe shortcut. Removing B1 can expose text in the next column or affect dependencies and row sizing.

Abbreviated XML. Style IDs are local.

## 24 Metadata has its own predicates

**Source**

```
- <definedName name="Print_Area"
-  localSheetId="0">S!$A$1</definedName>
  <definedName name="_xlnm.Print_Area"
   localSheetId="0">S!$A$1</definedName>
```

**Guarded result**

```
  <definedName name="_xlnm.Print_Area"
   localSheetId="0">S!$A$1</definedName>
```

Match a built-in alias and its prefixed counterpart in the same raw local scope. Keep names on different sheets distinct.

Unsafe shortcut. The current predicate does not compare formula bodies. This example must not imply general semantic equivalence.

Abbreviated XML. Style IDs are local.

# Guarded XML transformations

Each panel is 252 pt wide. Code is 8.2 pt at this scale.

## 25 Column widths position repeated borders

**Source**

```
<col min="8" max="8" width="4"/>
<col min="9" max="9" width="20"/>
<row r="2" s="1" customFormat="1"/>
```

=

**Retained by the guard**

```
<col min="8" max="8" width="4"/>
<col min="9" max="9" width="20"/>
<row r="2" s="1" customFormat="1"/>
```

A surviving row border repeats at column boundaries. Preserve existing widths even at empty positions.

Unsafe shortcut. Merging the empty columns moves the repeated border boundaries.

Abbreviated XML. Style IDs are local.

## 26 Compare font records, not just IDs

**Source**

```
 <c r="A1" s="1"><v>7</v></c>
 <c r="B1" s="1"/>
- <c r="C1" s="1"/>
 <!-- column uses another font ID -->
```

**Guarded result**

```
 <c r="A1" s="1"><v>7</v></c>
 <c r="B1" s="1"/>
```

Byte-identical font records share an internal identity. Retain the required representative without preserving duplicates merely for different font IDs.

Unsafe shortcut. Different font records retain their independent masks. Other XF components still matter.

Abbreviated XML. Style IDs are local.

## 27 Unsupported input is an exclusion

**Source**

```
<worksheet xmlns:s="...">
 <s:sheetData>...</s:sheetData>
</worksheet>
```

**Decision**

```
<!-- Excluded with part and reason. -->
<!-- No cleaned workbook published. -->
```

The fast parser targets unprefixed Transitional SpreadsheetML. Check bounded root headers, then namespace declarations during the normal stream.

Unsafe shortcut. An unsupported or malformed representation cannot be treated as empty and rewritten.

Abbreviated XML. Style IDs are local.

## 28 Prune stale chain entries by sheet

**Source**

```
 <calcChain>
  <c r="A1" i="1"/>
- <c r="C1" i="1"/>
  <c r="C1" i="2"/>
 </calcChain>
```

**Guarded result**

```
 <calcChain>
  <c r="A1" i="1"/>
  <c r="C1" i="2"/>
 </calcChain>
```

Only sheet 1 was processed and its C1 record was removed. Resolve chain sheet IDs through workbook relationships.

Unsafe shortcut. Do not delete a same-coordinate entry on an unprocessed sheet or when its mapping is unknown.

Abbreviated XML. Style IDs are local.

# One representative, one native layout effect

Reduced fixture | installed Excel 16.0 build 20326 | full rebuild before each capture

### One font witness preserves row layout

A2 has Arial 10. Empty B2 and C2 have Arial 18.

**Source**

```
  <row r="2">
   <c r="A2">...</c>
   <c r="B2" s="1"/>
-  <c r="C2" s="1"/>
  </row>
```

**Guarded result**

```
  <row r="2">
   <c r="A2">...</c>
   <c r="B2" s="1"/>
  </row>
```

**Original control** **23.25 pt**

| | | | |
|---|---|---|---|
| Font witness control | | | |
| 10 pt text | | | |
| Next visible row | | | |
| | | | |

**One witness retained** **23.25 pt**

| | | | |
|---|---|---|---|
| Font witness control | | | |
| 10 pt text | | | |
| Next visible row | | | |
| | | | |

**Both witnesses removed** **12.75 pt**

| | | | |
|---|---|---|---|
| Font witness control | | | |
| 10 pt text | | | |
| Next visible row | | | |
| | | | |

Native Excel after full rebuild. Removing the repeated witness is safe here. Removing both moves the next visible row.

### What was changed

The three input packages differ only in sheet1.xml. The source has two empty Arial 18 cells. The one-witness package removes C2. The negative control removes B2 as well. All use the same fonts, text and columns.

### What Excel rendered

The source and one-witness captures have identical decoded pixels. Both have row 2 height 23.25 pt. The negative control has row 2 height 12.75 pt and moves the next visible row upward. Its image is shorter.

### Scope

These are constructed native controls, not the Kulturfinanzbericht screenshots. They illustrate the font-witness mechanism independently of the documented corpus case. They are not a full validation run or a measurement of shrink throughput.

# A smaller file can expose a different fill

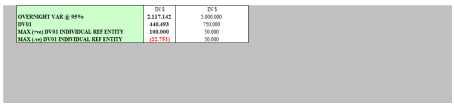
Retained Enron case ccde4294bab9094d7d73 | PL-Consolidated!A25:H37

## An empty gray cell masks an inherited pink fill

Column B uses XF 51 (pink). Cell B31 uses XF 98 (gray).

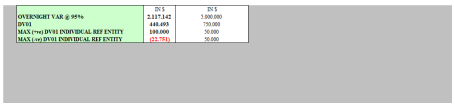
### Original

```
<c r="B31" s="98"/>
```



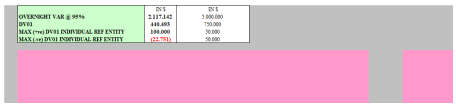
### SheetShrink Default

```
<c r="B31" s="98"/>
```



### Excel Check Performance

```
<!-- B31 removed -->
```



PL-Consolidated!A25:H37 after full rebuild and Save. Original and Default match pixel-for-pixel here. Removing the empty gray cells exposes the inherited pink background.

### Original, rebuilt

| | | |
|---|---|---|
| **OVERNIGHT VAR @ 95%** | IN $<br>**2.117.142** | IN $<br>5.000.000 |
| **DV01** | **440.493** | 750.000 |
| **MAX (+ve) DV01 INDIVIDUAL REF ENTITY** | **100.000** | 50.000 |
| **MAX (-ve) DV01 INDIVIDUAL REF ENTITY** | **(22.751)** | 50.000 |

### Default, rebuilt

| | | |
|---|---|---|
| **OVERNIGHT VAR @ 95%** | IN $<br>**2.117.142** | IN $<br>5.000.000 |
| **DV01** | **440.493** | 750.000 |
| **MAX (+ve) DV01 INDIVIDUAL REF ENTITY** | **100.000** | 50.000 |
| **MAX (-ve) DV01 INDIVIDUAL REF ENTITY** | **(22.751)** | 50.000 |

### Check Performance, rebuilt

| | | |
|---|---|---|
| **OVERNIGHT VAR @ 95%** | IN $<br>**2.117.142** | IN $<br>5.000.000 |
| **DV01** | **440.493** | 750.000 |
| **MAX (+ve) DV01 INDIVIDUAL REF ENTITY** | **100.000** | 50.000 |
| **MAX (-ve) DV01 INDIVIDUAL REF ENTITY** | **(22.751)** | 50.000 |

# What the rejected shortcuts would change

Schematic illustrations | these are not Excel screenshots or measured coordinates

## 11 Contiguous centering span

**Keep**




**Unsafe shortcut**




One surviving blank cannot replace the complete span.

## 04 Local no-fill mask

**Keep**




**Unsafe shortcut**




Removing the K-cell mask exposes its blue parent fill.

## 25 Empty column boundaries

**Keep**



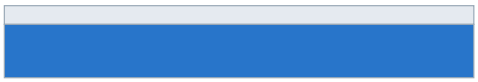
**Unsafe shortcut**



Changing widths moves borders even without values.

## 13 Visible exterior tail

**Keep**



**Unsafe shortcut**



A blue tail over white columns remains visible.

# Source index and evidence limits

Pinned implementation e6ae437a2d72 | full paths, anchors and hashes are in cases.json

### 01-content  Content is not formatting

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | preserves_value_and_inline_string_elements_with_attributes

S4. Content tags are matched at their element-name boundary.

### 02-font-witness  Keep one font witness

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | automatic_height_keeps_font_without_apply_font_or_row_style

Native control uses a reduced 18 pt fixture. The corpus regression is Kulturfinanzbericht, Impressum!B16, Arial MT 12 and no applyFont flag. Its documented heights are 15 and 12.75 pt.

### 03-merged-font  A witness is local to a merge

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | merged_text_keeps_one_local_font_even_without_apply_font

Destatis Informationen zur Statistik!A6:H6. Native controls reported 43,767 changed pixels with all blanks removed, zero with B6 retained. Original images are retained on another machine.

### 04-no-fill  No fill can be a visible override

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | unstyled_visual_masks_are_not_tentative_font_sentinels

PnP Sector Boilers, Fig 4 - SCR. An explicit XF with applyFill=1 and fillId=0 has the same guarded masking role.

### 05-redundant-no-fill  Remove an override with nothing to clear

excel-shrink-core/src/sheet/tail_fill_regression_tests.rs

regression test | enron_august_no_fill_overrides_do_not_anchor_an_empty_sheet

Enron AUGUST, bundle 44ad86f1d9f249470fc6. The corpus case retained 21,947 unnecessary overrides before the fix.

### 06-row-xf-zero  An implicit row style still overrides

excel-shrink-core/src/sheet/cells_regression_tests.rs

regression test | custom_format_without_style_uses_default_row_style

Raw Data F187:F190 is an independently restored Enron control. The AFFF manual Data rows 124-128 motivated the current row-format guard.

# Source index and evidence limits

Pinned implementation e6ae437a2d72 | full paths, anchors and hashes are in cases.json

### 07-column-font-mask  Font witnesses cannot replace local masks

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | automatic_row_keeps_each_mask_of_an_unrepresented_hidden_column_font

Data.gov LCOE model, Installed Cost Overview. Documented row 14 changed from 14.25 to 15 pt before the correction.

### 08-row-promotion  Move repeated formatting into the row

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | dense_full_width_row_compacts_identical_empty_styles_into_row_header

The full-width test removes 16,383 empty cells. No new style index is allocated.

### 09-promotion-gap  Restore necessary exceptions at gaps

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | dense_row_preserves_internal_and_trailing_column_styles_on_repeat_cleanup

The regression uses gaps at E1, AD1, XFC1 and XFD1. It also checks a second cleanup pass.

### 10-automatic-promotion  A cached height is not a fixed height

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | dense_row_promotion_requires_an_explicit_fixed_height

verkehrsunfaelle-zeitreihen, 3.8_(1). Native repair matched the 1,358 by 1,327 original capture. This drawing is a guard summary, not exact output XML. The excerpt shows the absence of row promotion. It does not enumerate ordinary cell cleanup.

### 11-center-span  A centering span needs every blank

excel-shrink-core/src/sheet/output.rs

regression test | centered_text_keeps_only_its_contiguous_alignment_span

Data.gov Fig 10_New 10-29-19. The native counterfactual restored D2:G2, not a single sentinel.

### 12-protected-font  Protection does not erase a font effect

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | exterior_protection_style_can_still_supply_a_required_font_sentinel

Enron 005d034f18d6dc340a2c had five one-pixel row-height changes. Active and inactive protection are covered by focused controls.

# Source index and evidence limits

Pinned implementation e6ae437a2d72 | full paths, anchors and hashes are in cases.json

### 13-white-tail  Prune only a genuinely redundant fill

excel-shrink-core/src/sheet/tail_fill_regression_tests.rs

regression test | redundant_white_fill_and_font_override_do_not_anchor_an_empty_tail

Two Enron Feb cases ended at rows 486 and 485 after cleanup. Native white and blue controls matched selected original regions.

### 14-content-free  A font alone need not anchor a tail

excel-shrink-core/src/sheet/tail_fill_regression_tests.rs

regression test | enron_nov_styleless_representatives_do_not_anchor_a_content_free_sheet

Enron NOV, bundles 44ad86f1d9f249470fc6 and a8bd950dff33454721b5. Exterior blank-grid geometry is outside this pruning contract.

### 15-border-mask  A row header can clear column borders

excel-shrink-core/src/sheet/tail_fill_regression_tests.rs

regression test | borderless_tail_row_masks_a_visible_column_border_without_retaining_cells

Santavy Table Sup E motivated the initial-tail-range guard. No full row of cells is added.

### 16-border-height  Empty row heights position visible borders

excel-shrink-core/src/sheet/output.rs

regression test | inherited_column_borders_keep_visible_tail_heights_only

The ordinary tail scan and content-boundary shortcut share this guard. Heightless and hidden redundant headers retain normal pruning.

### 17-conditional-layout  Conditional ranges protect row layout

excel-shrink-core/src/sheet/tail_fill_regression_tests.rs

regression test | conditional_regions_preserve_empty_row_heights_without_restoring_cells

RN17B-RN30 Volatiles. Unknown conditional scope conservatively protects existing layout through the final Excel row.

### 18-drawing-anchor  Objects extend the layout boundary

excel-shrink-core/src/sheet/output.rs

regression test | drawing_text_box_anchor_preserves_layout_row_after_last_cell

Missing or unsupported drawing information is a remaining review concern, not proof that no object exists.

# Source index and evidence limits

Pinned implementation e6ae437a2d72 | full paths, anchors and hashes are in cases.json

### 19-column-font  Merge tail columns with font witnesses

excel-shrink-core/src/sheet/cols.rs

implementation anchor | tail_font_sentinels

Unknown styles, overlap, hidden columns, borders, uncovered gaps and crossing data boundaries restrict merging. Visible row borders also expose column widths.

### 20-hidden-tail  Hidden constants need deferred approval

excel-shrink-core/src/sheet/hidden_content.rs

implementation anchor | MIN_TAIL_CELLS

4,096 is a conservative admission threshold, not proof of user intent. Deferred deletion re-cleans touched rows and invalidates stale tail anchors.

### 21-shared-formula  Translate shared references by axis

excel-shrink-core/src/reference_scanner_sheet_scope_tests.rs

regression test | shared_formula_protects_shifted_hidden_inputs

Destatis verkehr-aktuell, 1.1.5. Unresolved groups, dynamic references or unknown scope disable reference-sensitive deletion.

### 22-chart-format  A blank chart-source cell carries format

excel-shrink-core/src/reference_scanner/object_references.rs

regression test | chart_format_anchors_keep_only_the_first_source_cell

Current format-anchor discovery resolves direct worksheet references, not named or dynamic chart sources.

### 23-whitespace  Whitespace can block text overflow

excel-shrink-core/src/shared_strings_tests.rs

regression test | detects_rich_text_shared_string_as_whitespace_only_when_all_runs_are_whitespace

Optional profiles have separate reference, protection and layout guards. ASCII whitespace classification does not cover every Unicode representation.

### 24-metadata  Metadata has its own predicates

excel-shrink-core/src/workbook.rs

regression test | equivalent_alias_scope_is_independent_of_attribute_order_and_quotes

Calculation-chain cleanup separately removes entries for removed cells only when sheet mappings resolve. Unaffected styles, shared strings and binary assets retain their indices and content.

# Source index and evidence limits

Pinned implementation e6ae437a2d72 | full paths, anchors and hashes are in cases.json

### 25-border-width  Column widths position repeated borders

excel-shrink-core/src/sheet/output.rs

regression test | visible_row_borders_preserve_empty_column_widths

This guard adds no cells and is propagated across parallel sheet chunks.

### 26-font-identity  Compare font records, not just IDs

excel-shrink-core/src/sheet/rows_regression_tests.rs

regression test | removes_repeated_empty_cells_when_column_font_records_are_identical

Style and font indices in the package are not renumbered. No general visual-equivalence test is inferred.

### 27-eligibility  Unsupported input is an exclusion

excel-shrink-cli/src/pipeline_namespace_tests.rs

regression test | late_namespaces_abort_output_without_replacing_prior_results

Ordinary default namespaces and drawing extension prefixes are allowed. Strict SpreadsheetML, UTF-16 and explicit core prefixes have separate exclusion guards. The namespace URI is abbreviated in the drawing.

### 28-calculation-chain  Prune stale chain entries by sheet

excel-shrink-cli/src/pipeline_regression_tests.rs

regression test | calc_chain_removes_missing_cells_for_processed_sheet_only

This is structural chain maintenance. Formula evaluation and replacement calculation caches remain Excel operations.